# Sheet 1

*Lectures 1 & 2 — Questions*

I. **Part A — Multiple Choice.** 77 questions on language levels, number systems, registers and memory addressing.

II. **Part B — True or False.** 50 statements.

**Notation.** [L1, S5] means Lecture 1, slide 5. [L2, S39] means Lecture 2, slide 39.
*[Section note]* marks material that the sheet uses but Lectures 1–2 do not cover.

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# Part A · Multiple Choice

*Part A · Multiple Choice (77 questions)* Name: ..... ID: .....

*Choose the one best answer for each question.*

**1.** What is Assembly Language?
(a) A high-level programming language (b) A low-level programming language
(c) A database language (d) A markup language

**2.** Assembly language is mainly used to communicate with the:
(a) Compiler (b) Operating system only
(c) Computer processor (d) Database

**3.** Which language is closest to machine language?
(a) Java (b) Python
(c) Assembly Language (d) SQL

**4.** Machine language consists of:
(a) English words (b) Binary instructions
(c) Assembly mnemonics (d) Flowcharts

**5.** An assembly language program is converted into machine code by an:
(a) Interpreter (b) Assembler (c) Editor (d) Linker only

**6.** Which of the following is an example of an assembly mnemonic?
(a) MOV (b) PRINT (c) HTML (d) SELECT

**7.** Which component executes machine instructions?
(a) CPU (b) Keyboard (c) Monitor (d) Printer

**8.** Which number system is commonly used to represent machine-level data?
(a) Binary (b) Decimal only (c) Roman (d) Alphabetic

**9.** The hexadecimal number system has a base of:
(a) 2 (b) 8 (c) 10 (d) 16

**10.** The binary number system has a base of:
(a) 2 (b) 8 (c) 10 (d) 16

**11.** The decimal number 10 is represented in hexadecimal as:
(a) A (b) B (c) 10 (d) F

**12.** Which register is commonly used as an accumulator in many architectures?
(a) AX (b) IP (c) SP (d) CS

**13.** In x86 assembly, AX is a:
(a) 8-bit register (b) 16-bit register
(c) 32-bit register only (d) 64-bit register only

**14.** Assembly language is considered machine-dependent because:
(a) It does not use instructions
(b) Its instructions depend on the processor architecture
(c) It can only run on the internet
(d) It requires no CPU

**15.** Which of the following is a common advantage of assembly language?
(a) Very high-level abstraction (b) Direct control over hardware resources
(c) No knowledge of hardware is required (d) It is always easier than Python

**16.** One disadvantage of assembly language is that it is generally:
(a) Machine-independent (b) Difficult to write and maintain
(c) Unable to perform calculations (d) Not executable

**17.** The assembler converts:
(a) Machine code into English
(b) Assembly language into machine/object code
(c) Python into Java
(d) Binary into decimal only

**18.** What is the hexadecimal representation of binary 1111?
(a) 10 (b) E (c) F (d) 15

**19.** The main purpose of learning Assembly Language is to understand:
(a) How high-level websites are designed
(b) How software interacts closely with the computer hardware and CPU
(c) How databases store tables
(d) How to create presentations

**20.** Which stage combines object files and libraries to create an executable program?
(a) Linking (b) Editing (c) Scanning (d) Compiling only

**21.** What does a mnemonic represent in Assembly Language?
(a) A memory address
(b) A symbolic representation of a machine instruction
(c) A hardware device
(d) A variable

**22.** Which of the following is a valid x86 register?
(a) AX (b) RX (c) ZX (d) QX

**23.** Which register is commonly used as a counter in x86 instructions?
(a) CX (b) IP (c) SP (d) CS

**24.** CX traditionally stands for:
(a) Code Extension (b) Count Register
(c) Control Extension (d) Computer Register

**25.** How many bits are in a 64-bit register?
(a) 16 (b) 32 (c) 64 (d) 128

**26.** How many general-purpose 16-bit registers are commonly associated with the 8086?
(a) 2 (b) 4 (c) 8 (d) 16

**27.** Which register is commonly used as an accumulator in the 8086?
(a) AX (b) BX (c) CX (d) DX

**28.** Which register is commonly used as a base register?
(a) AX (b) BX (c) CX (d) DX

**29.** Which register is commonly used as a counter?
(a) AX (b) BX (c) CX (d) DX

**30.** Which register is commonly used for data and I/O operations?
(a) AX (b) BX (c) CX (d) DX

**31.** Which segment register is associated with the stack?
(a) CS (b) DS (c) ES (d) SS

**32.** What is the size of an 8086 address bus?
(a) 8 bits (b) 16 bits (c) 20 bits (d) 32 bits

**33.** What is the maximum memory address space of the 8086?
(a) 64 KB (b) 128 KB (c) 512 KB (d) 1 MB

**34.** Which register is divided into AH and AL?
(a) AX (b) BX (c) CX (d) DX

**35.** How many general-purpose 16-bit registers does the 8086 have?
(a) 4 (b) 6 (c) 8 (d) 10

**36.** Which register is known as the accumulator?
(a) BX (b) AX (c) CX (d) DX

**37.** What is the size of the AX register in the 8086?
(a) 4 bits (b) 8 bits (c) 16 bits (d) 32 bits

**38.** Which two 8-bit registers make up AX?
(a) BL and BH (b) CL and CH (c) AL and AH (d) DL and DH

**39.** What is the lower 8-bit portion of AX called?
(a) AH (b) AL (c) AXL (d) AXH

**40.** What is the higher 8-bit portion of AX called?
(a) AH (b) AL (c) AXH (d) AHH

**41.** Which register is commonly used for multiplication and division operations?
(a) AX (b) BX (c) CX (d) SP

**42.** Which register is commonly used for I/O operations?
(a) AX (b) BX (c) CX (d) DX

**43.** Which register is divided into BH and BL?
(a) AX (b) BX (c) CX (d) DX

**44.** How many segment registers does the 8086 have?
(a) 2 (b) 3 (c) 4 (d) 6

**45.** Which register contains the address of the code segment?
(a) DS (b) SS (c) ES (d) CS

**46.** Which segment register is used for data?
(a) CS (b) DS (c) SS (d) ES

**47.** Which segment register is associated with the stack?
(a) CS (b) DS (c) SS (d) ES

**48.** How does the 8086 calculate a physical memory address?
(a) Segment + Offset (b) Segment × 10H + Offset
(c) Segment × Offset (d) Segment – Offset

**49.** If CS = 2000H and IP = 0100H, what is the physical address?
(a) 2010H (b) 2100H (c) 20100H (d) 20010H

**50.** If DS = 1000H and the offset is 0020H, what is the physical address?
(a) 10020H (b) 10200H (c) 10002H (d) 12000H

**51.** What is the size of the 8086 data bus?
(a) 8 bits (b) 16 bits (c) 20 bits (d) 32 bits

**52.** Which of the following is NOT a general-purpose register in the 8086?
(a) AX (b) BX (c) CS (d) DX

**53.** Which register is most commonly associated with arithmetic operations?
(a) AX (b) CS (c) SP (d) IP

**54.** What is the full 16-bit register containing AH and AL?
(a) BX (b) AX (c) CX (d) DX

**55.** If AL contains 25H, which part of AX does it represent?
(a) High byte (b) Low byte (c) High word (d) Segment

**56.** If AH = 12H and AL = 34H, what is AX?
(a) 1234H (b) 3412H (c) 1243H (d) 4321H

**57.** Which register is normally used for loop counting?
(a) AX (b) BX (c) CX (d) DX

**58.** Which segment register is commonly used as an additional data segment?
(a) ES (b) SS (c) CS (d) IP

**59.** What is the physical address of 0000H:0000H?
(a) 00001H (b) 00010H (c) 00000H (d) 10000H

**60.** What is the physical address of 1000H:0000H?
(a) 01000H (b) 10000H (c) 10001H (d) 11000H

**61.** What is the physical address of 1000H:0010H?
(a) 10010H (b) 10001H (c) 10100H (d) 11010H

**62.** What is the physical address of FFFFH:0000H?
(a) FFFF0H (b) FFFFFH (c) FFF00H (d) 10000H

**63.** What is the maximum physical address available in the 8086?
(a) FFFFH (b) FFFFFH (c) FFFFFFFH (d) 10FFFH

**64.** A native machine language is a language of level:
(a) L1 (b) L2 (c) L3 (d) L4

**65.** Each computer has a native machine language that runs directly on its:
(a) Operating system (b) Software
(c) Hardware (d) All above

**66.** A human-friendly language is of level:
(a) L1 (b) L2 (c) L3 (d) L4

**67.** A high-level language is of level:
(a) L1 (b) L2 (c) L3 (d) L4

**68.** Assembly language is of level:
(a) L1 (b) L2 (c) L3 (d) L4

**69.** Instruction set architecture (ISA) is of level:
(a) L1 (b) L2 (c) L3 (d) L4

**70.** Digital logic is of level:
(a) L1 (b) L2 (c) L3 (d) L4

**71.** Digital logic means:
(a) Level 1 (b) CPU
(c) Implemented using bipolar transistors (d) All above

**72.** The binary conversion of the decimal number $(21)_{10}$ is:
(a) $(00010101)_2$ (b) $(00010110)_2$ (c) $(00011001)_2$ (d) $(10010101)_2$

**73.** $(00000100)_2 + (00000111)_2 =$
(a) $(00010111)_2$ (b) $(00010101)_2$ (c) $(00001011)_2$ (d) $(10011101)_2$

**74.** The largest unsigned integer that may be stored in 20 bits is:
(a) 1048576 (b) 1048575 (c) 1048577 (d) 1048557

**75.** Convert $(111110011100)_2$ to hexadecimal:
(a) $(F9A)_{16}$ (b) $(FAB)_{16}$ (c) $(E9C)_{16}$ (d) $(F9C)_{16}$

**76.** $(6A)_{16} + (4B)_{16} =$
(a) $(B5)_{16}$ (b) $(A3)_{16}$ (c) $(C5)_{16}$ (d) $(E3)_{16}$

**77.** $00001100_2 - 00000011_2 =$
(a) 00001001 (b) 00001101 (c) 00001011 (d) 00011001

# Part B · True or False

*Part B · True or False (50 statements)* Name: ..... ID: .....

*Circle **T** if the statement is true and **F** if it is false.*

**1.** A byte consists of 16 bits. (T) (F)

**2.** The binary number 1111 is equal to hexadecimal F. (T) (F)

**3.** 1 hexadecimal digit = 8 bits. (T) (F)

**4.** 1 hexadecimal digit = 4 bits. (T) (F)

**5.** 2 hexadecimal digits = 1 byte. (T) (F)

**6.** 1111b = Fh. (T) (F)

**7.** 1010b = Ah. (T) (F)

**8.** 11111111b = FEh. (T) (F)

**9.** 11111111b = FFh. (T) (F)

**10.** Hexadecimal 10h is equal to decimal 10. (T) (F)

**11.** Hexadecimal 20h is equal to decimal 32. (T) (F)

**12.** 0Ah is equal to decimal 10. (T) (F)

**13.** 1010b is equal to hexadecimal Ah. (T) (F)

**14.** Decimal 15 is represented as 0Fh in hexadecimal. (T) (F)

**15.** The largest unsigned value that can be stored in 8 bits is 255. (T) (F)

**16.** An assembler translates Assembly Language instructions into machine code. (T) (F)

**17.** Assembly language is completely independent of the processor architecture. (T) (F)

**18.** MASM stands for Microsoft Macro Assembler. (T) (F)

**19.** The binary number 101101b is equal to decimal 45. (T) (F)

**20.** The hexadecimal number 2Ah is equal to decimal 42. (T) (F)

**21.** Hexadecimal FFh requires 8 bits to represent. (T) (F)

**22.** BYTE occupies 1 byte. (T) (F)

**23.** WORD occupies 2 bytes. (T) (F)

**24.** DWORD occupies 4 bytes. (T) (F)

**25.** QWORD occupies 8 bytes. (T) (F)

**26.** DWORD is 16 bits. (T) (F)

**27.** DWORD = 32 bits. (T) (F)

**28.** A BYTE occupies 8 bits in memory. (T) (F)

**29.** A WORD occupies 4 bytes. (T) (F)

**30.** A DWORD occupies 4 bytes. (T) (F)

**31.** AX is a 32-bit register. (T) (F)

**32.** EAX is a 32-bit register. (T) (F)

**33.** AL is the low 8 bits of AX. (T) (F)

**34.** AX = AH + AL. (T) (F)

**35.** AH contains the low 8 bits of AX. (T) (F)

**36.** One hexadecimal digit represents 4 bits. (T) (F)

**37.** Two hexadecimal digits represent one byte. (T) (F)

**38.** 1000h and 1000 always represent the same value. (T) (F)

**39.** BYTE 10,20,30 reserves 3 bytes. (T) (F)

**40.** WORD 10,20,30 reserves 3 bytes. (T) (F)

**41.** DWORD 1,2,3 occupies 12 bytes. (T) (F)

**42.** Executing a machine instruction requires three basic steps: fetch, decode, and execute. (T) (F)

**43.** Assembler is a utility program that combines individual files into a single executable program. (T) (F)

**44.** Each assembly language instruction corresponds to a single machine language instruction. (T) (F)

**45.** Memory access takes more machine cycles than register access. (T) (F)

**46.** The Real-address mode can address a range of 4 MB. (T) (F)

**47.** The hexadecimal form of the binary number 00100111 is 27. (T) (F)

**48.** A computer system usually contains four bus types: data, I/O, control, and address. (T) (F)

**49.** "Cache miss" describes when data to be read is already in cache memory. (T) (F)

**50.** "Cache hit" describes when data to be read is already in cache memory. (T) (F)

*— End of Sheet 1 —*